

4. A ball of mass 3 kg has 750 J of potential energy when held at the top of a building. The ball is then dropped from rest.

- (a) (i) Use an equation from page 2 to calculate the height,  $h$ , of the building. (acceleration due to gravity,  $g = 10 \text{ m/s}^2$ ) [3]

height,  $h = \dots\dots\dots \text{ m}$

- (ii) When the ball hits the ground it has 600 J of kinetic energy. Use an equation from page 2 to calculate the velocity of the ball at this moment. [3]

velocity =  $\dots\dots\dots \text{ m/s}$

- (iii) Use the information above to calculate the work done against air resistance as the ball falls. [1]

work done =  $\dots\dots\dots \text{ J}$

- (iv) Use an equation from page 2 and your answers to parts (a)(i) and (iii) to calculate the mean force of air resistance during the fall. [3]

mean force =  $\dots\dots\dots \text{ N}$



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(b) (i) Explain, in terms of named forces, why the ball reaches a terminal speed as it falls through the air. [3]

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(ii) Newton’s 3rd law of motion applies to the ball as it falls.  
**Complete the table** to describe the Newton’s 3rd law force for each of the actions given. [2]

| Action                                | Newton’s 3rd law force |
|---------------------------------------|------------------------|
| Earth pulls downwards on the ball     | .....                  |
| Ball pushes downwards against the air | .....                  |

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